Manage Azure Key Vaults

ONTAP 9.14.1 REST API reference

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Manage Azure Key Vaults

Security azure-key-vaults endpoint overview

Overview

Azure Key Vault (AKV) is a cloud key management service (KMS) that provides a secure store for secrets. This feature allows ONTAP to securely store its encryption keys using AKV. In order to use AKV with ONTAP, you must first deploy an Azure application with the appropriate access to an AKV and then provide ONTAP with the necessary details, such as key vault name, application ID so that ONTAP can communicate with the deployed Azure application. The properties "state", "azure_reachability" and "ekmip_reachability" are considered advanced properties and are populated only when explicitly requested.

Examples

Enabling an AKV configuration for an SVM using the certificate authentication method

The example AKV configuration is enabled for a specific SVM. Note the *return_records=true* query parameter is used to obtain the newly created key-manager keystore configuration.

```
# The API:
POST /api/security/azure-key-vaults

# The call:
curl -X POST 'https://<mgmt-ip>/api/security/azure-key-
vaults?return_records=true' -H 'accept: application/hal+json' -d "{
\"svm\": { \"uuid\": \"4f7abf4c-9a07-11ea-8d52-005056bbeba5\" },
\"client_id\": \"client1\", \"tenant_id\": \"tenant1\", \"name\":
\"https:://mykeyvault.azure.vault.net/\", \"key_id\": \"https://keyvault-
test.vault.azure.net/keys/key1/a8e619fd8f234db3b0b95c59540e2a74\",
\"client_certificate\" :
\"MIIQKQIBAzCCD+8GCSqGSIb3DQEHAaCCD+AEgg/cMIIP2DCCBg8GCSqGSIb3DQEHBqCCBgAw
ggX8AgEAMIIF9QYJKoZIhvcNAQcBMBwGCiqGSIb3DQEMAQYwDgQIWkY7ojViJDYCAggAgIIFyJ
PjIfmM6yTCKVw5ep2oZLwwvRca8pKhISVjw+WjWngh/f6Py/Ty0CwCjDFUZPsUUdSmk78E7SAz
0CpQyBwmUuFJQShjZjftHLKRWld3O4sJKB8DzH9Yw1C7En94cyJ1rT4WYoVFmeJcmOXx6h+NFH
c7njtXVsKwxc5BF88K3+3kHdV3WyVdXoeXe7yY/+EjFfjtBryp8ljuielX/NFlh5kowhoj+yxn
O0c1/0OI1iV3mTIOTXD8qrZVp9ZhAxSTRBd5uDyWMfppqxW2L+9vCUU+ZgmRxtU3VsRLOp/T14
0OP7Sn1Ch2OE0bIrbYYtcpi04QcUtfEJBMlbbTbJPHDAtiO2KIQKviZL4QMZgho9NNgL4MUpIb
NSzDCbuIC+nNMXfgfs0nPZewY+b43H/tMmnZ8Q4kiCFwrUqbFbflBiPMOaJsS0eQaJhDmzM90Q
EgbesHWgPreAcfMUcN1+BaqHFLHUxLXDxQix6zYiCAtDX6/EKlirRh1TFpmFX2PBd+X6uODhmw
m4ub9RKj3In8t5qgtN4q/mTBXjAVDAbTIIEgobBRaXGSSXCBc9W/jRed0DRZD9Bm8T/nV39sZN
ducwZa5ojYTX8fFMA0cfY6IFivXHjB00coHEEGdgCfC0G8vACqLbb+2NuhMJPtR7Ig50iAPUMc
670Z5ItOTQhyYOZ/KagOtvV8sKPCzeAkcMoHlsml89V79zt1fCJQTVWnaGiMj5Orcbskk6vCxh
DGeU6q1kgvXJKXOYRF8/wIpv8Y7/rEpnGwE/I0ZOXzdIDHXqA53B1zyOVem25ezWCD+kpoH89X
JssYlNjIMJhjVRED61w/DbSXg2yFu/v3ckGapVvTuyAiz5hWUNfl3pt++da6GoekKnLqtL4G/R
GXCnebLbXg838dlTGBznoCwGTVxXDeVYafz8AjI10qYtTMcbN56ya9kK7IHSkrnFX24xQRQOfm
D0Vob71pjdz8r1aXKvD/1X2TkYJHoeEHq0nWpU8vwDG/xhv4YgKJGN9qsEZgiTXETUh5gak8e1
```

tGNkP+fum+1OqlO5oS+SwNa5/eB8eFeJl2Oi48Xi5UapaTRHPFp6kZfPXOu9cEjhILowRIi6gl
g7FUbmoJcu5OvDIyP9JlyQklw2VtgNlm1QOIvzRenXmy18XnP50NTxx2cIwby8tIcdSn2C2qhj
8Gk7q8oxVZGiBgtz4BwyzyKkypwm60BBRrHpAKLw6JM5RISeZnYQfIsId0tGgb61go0RJf0sFt
buvZcSvLI+2Onj8KH1TlmMR4dbuCWE9Ym4sVRmD1D6/f6BoNH0DRg7TJkEFbOadJsNPGzHbKte
LdaSMGTNUZ3hEDQeomakQMfvCgypbOLxrTTqfbenHRtN+iFNYW0zCUW6EJoAXp+lqFnwQL52Il
2QxwZikE01P2k0GharzAJkXnNaFGnmHIIP6wJrCCSDZwDmr7GI2R5evDlRi17QUg2sulxQV0U8
zezzwIUgEe/Whf0ngGJv/QcsL2jyri/tSQbUWs4g+yep4SlE3iddhfqSJzI2iKdAE+HLiHGVO1
z70fGEsO6dPLnmh4eoWidgZi9N/SoBy1aT0JpIQ6z6N5ImPfDWu9Y6TWXUg1iyOIXGsxIQVIgU
NoB5Ru/ApDxpYpFLk0fH9k9OnEWK5Im33puOQKLno1uwrOmdbG8+x1EY8wc9FvkHGH0Zh4Hydi
CVUcYSdiGWUxVmgm4OgyiYzcpB+Ar2dzikGc4pBg8fa1a1HN5Q3TK3w4h/HeOUlmA4vWOYuVO1
H93ILGP6PWfkug+1Tam6+8yD0W5meiZ0UIZR8TF/9gDb4+4wTFnPwgfTrggEauA8tt8uJtiyBC
rYexgZTXIZGTUj/86KXQaJKCreRr/kqwJOWqkNW4CGUVzw7LiI+sArOZqUp/TsxnbNC73XCMNl
PsnByb2zCeK13V26Crl84U9sDuqQTJRaIse01MN9AAjpa2QWEwggnBBgkqhkiG9w0BBwGgggmy
BIIJrjCCCaowggmmBgsqhkiG9w0BDAoBAqCCCW4wgglqMBwGCiqGSIb3DQEMAQMwDgQIEjm88b
1+pnkCAggABIIJSDD3P+vnllSo1mQvmYgZVfV37T3KpurJvMxQScPvalWiF7Q1Iwasf/+N0hKK
Nr2j/aGZLunLkaG6mLPeBP2l2LCwnUxDu5kYffVVE90WX/bXewbYQribwFNkNhUrSgen8BfhnR
lvDrzbBLoHIvDrUFszSVBCYh31Vwgu8p9SjC8K/XlumcLdjSFko85XpoK23euhowjWH+X0kRoY
GzorcdNE8z03BKvfR61W2XWzTSaWQ6eZHGs6Urnx5Fe/w50U9tMIi3BCCCqgapUHVdmHqKkmWL
ikX8LssUcN30JVekM2aJ9v4YO6CoegKAMVDs0tVSOv3KbGC3GNX6lgHu4y1LOZPlPLfPXb0wDH
qavlxK3zpHl8sIRzuX3HXSdEdenHYAkSV/IQZ89h+CZUkf0nu/og8eoA8ATDA5g7fj3HXpQ6cY
drUBaHc7ruxHOiWR0GcT4XK4TTz7zZTO1wWPViprUo6ayw0dYZSG22MeDA027YirM044Ifosn9
CsqnNLZoOWvA2ao4ippDoBRqv5Hv6n0I3fOAys5nPq3jJtKQ5neqUYo0MrAkoKHo0h6zn0Bfvi
syB88aM9N0mPD76ykbAERq7151biKbA2tk8bb9dy/sJmk2ojM/D/W1YtrNL4iM6azL2kVN5eiC
hxCof33/RuRpXfGR8YNeJTl7bq42wL70QKDBRoG1TPcLqdVqz74oshlRspfqvZsbsUatbASBt2
T0YG4zfgfGh7sb2ezyougVvzdp77wAJ6n39dc/ZLDdYDzFkQb07984y8LlhIM1AcwFcMh43gWp
6A8CJ02l74ednirSqSVOPZ7K9dRw6Y0X8MB4/WGzEcvFeHYIGLBcXi1sBY5wjWnbeuh1wLiSkM
DQRB6oGOvF7bJsilkx5PwgWbbqw8KUSuU01skbMAa5T8Hkm4OiSTf2a78E0zIKLGZg7yu9FDII
tWYWOkG96MXEBAdOuH+wWYmaEexh51ONrfFwKDuDMZh7MO20TTEQU8oQdjRRoAofXvTcj22GSM
TY6XleskZX2ZKxSQdD1tCtkjGRKHSTYza3zLHbBiJTIJw4z6sw9FyTTApg66UAkNtiMa1r9nqT
TNaxRWEXMEQVRLzAL2F9aqjgW65xrbYXu/J9Y/SYTcYbX2SRA/JkQ+Y8F68KOoS1pvK1p5/FcE
DvprTNDS4lf+aj3HNWuK5wOsrpBhMlb2IfluK/9QwPh9IC/RhHRfimyTPRXAf73cehNdp8DpKw
Lm+jr30vazFwICpvSbi6Etb6GXfPkKaX7ztpQBqG92m2/0g3LWfPti1zwrPHPBz8y1qQMU268D
oo8YvWtI4KGaDAFb6XQhR6t6mqoq/3IP6/g//PZVEnSyUVsPLDJlLF9fiOwTbMZnaiscKv8SGE
s//B9JkKrdsRrQRZcnnPjJnJLILblRVAZGuXpSKSYVPzYmOjUx3sSeLSiPoSOcqRIJ0X3s4ED0
92W3tR4ZXK3fnkFyrIVtRJsB3k/2smiQ6Pc1VuKHh1yTzYjXKRQcDaY3EDP9IWFtjiUfZQoZci
jMWt6YXim23m2aN2Ed8qIedikR6OjFHE4Kus/2yegTszSs5CrM7NamKWzeIeNNth/cTcmT++GD
umsGNTBAsHHSq1KYpqLi4GKLHzU7WNCQRdAcIDEvMZH/CH1mZK7bzb9z038rPf/D5WZrcK1ttd
5BjTJjj7GerS0xLkvYIklAJqurjMdWYmQtT4JAHF90/zRKqFFVpSiW074bRQ+PfaLI5C+TwoX5
lYD+R91A0qyGKIkFITa8hZFY+Up+rSuREqnpAvdAVL9/gLPF6I+5+D+sVBsGRbw2rFVRbCHdwa
TQcAVPeJJy0f/+sOs/PXoejr3siORpf8iLLYOaziGYf1EtunFcCLj8PEOznaUyouJ+lm9YKPBS
LULC/sVVy6XUArYfJfq0Ag31YXpJeWPbORxVP/VCm8d/sNjWTQXGN/IjNZaZuliXNgq5nRkPBK
wF23ZUYG4pLGpGROLup9nLSgEbpiDmN1Gq/IHSfI/8HpG/yRAoCdqUdre3yL/f9caj8RBBHRYb
bfRxtyQ9u2vsrqo1oZ7F+Mu+kjuc9BxCMvJ7JaKwvQJckAkzTo6t10t6MzwiqJ7Au+2oOJ2Ukb
/985+TFGS219fmqWfwisOfpuvSkjRj8vIDBBm9itKIS+pVpfz+Mg7kl3WmkUrgF3yjTH5/C51u
aSzK2KeEVoWPx/Ps2CX7ATo6AsETp8Na38dT6d+Dm4WM4sBieKt/yOEFhiBNkgpVKAqawKRvLW
3U73OIKC8VLFhhnU+ogGxcUq5mZXvMbNDIaU2LvtmtPPo/qL0bOYu76TKc1ZX0R6AXkeImQgRP

```
sdeXPPANtW3la585oZbYxUXRfEIeKmkcv3eSGnPCVesbxxd1SaIJe2j7H9MbHdjYkeFQuECnUh
Kxg63BVPl/qAEIO5+OKBzM7ctuP8apeGW1iHAueKzJXc5IeFS/3iwkfDLRkrgzBeNIL0IINo3C
oGSvn95Z8+LhNSopyqt3uB4rQksUYIwXgkfrEVYujCO0T5dSkk5j10X7WlDm4DHZVLJH+GtL6v
9A6xFJNDQfQF0hS+wlXkTkMq7pUiX+Qohf8QRJZEyU5VWo2CesR63j1MFpkB3xybpbjt8oI47X
C20GEn3uCjwMwq/3K4ibHnqi16pPPRgI/u3R9TVfvOC2e0xgllrFG6cKUfogUaXoxHqP1KKjUw
23bpd9L09LzSDdSHcoDPokWzDee0ZP/Z6VH3rdjQR71kw4VBeT8nKfLP2dGBd0tpWDQhCFK7I9
axxxthnv0v09x/J7jhyoLRt5e8lMEfrqtnMWdqjFgYVEQndthZ+9/XvfNk6f5MD8fDheMuvbNT
hduFSZEcZCLlW4GWKneVji4wdBrV3aCrzAzxy0H7y7nnkyCEvac503UDtr1bk1VJIVsYfYrN2S
2DPbp3H2E8r/n6jfBilwFyp3JTJvnRqQTcYHXDieW8Njq46JO6O6wsPwKQTKMfHGxxTRJdRe5y
vJD54xvFWw1YEJ/Q2c8cr1NNXEN32e5psfIJ7o48k6bsiyXnbHKSjK781Z5h8Hc3FbUF2U2p5J
qLwcD7+bknEunsbWSC37iMk7oweF3hMhKRMm9iYJ8tpxMRcWCOt7ador+Y2fYWBsu/bwXwcRI0
8TElMCMGCSqGSIb3DQEJFTEWBBRymjnjEbJmrRwh4sRnwudfSQP6KDAxMCEwCQYFKw4DAhoFAA
QU+YFhgKEYjfXN/cL70yRrJSHFgUwECHeCTQnUEU0BAgIIAA==\" }"

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "024cd3cf-9a08-11ea-8d52-005056bbeba5",
    "svm": {
      "uuid": "4f7abf4c-9a07-11ea-8d52-005056bbeba5",
      "name": "vs0"
    },
    "client_id": "client1",
    "tenant_id": "tenant1",
    "name": "https:://mykeyvault.azure.vault.net/",
    "key_id": "https://keyvault-
test.vault.azure.net/keys/key1/a8e619fd8f234db3b0b95c59540e2a74",
    "_links": {
      "self": {
        "href": "/api/security/azure-key-vaults/024cd3cf-9a08-11ea-8d52-
005056bbeba5"
      }
    }
  }
]
}
```


Creating an inactive AKV configuration for an SVM using the client secret authentication method

The example AKV configuration is created for a specific SVM but is not enabled. Note the *create_inactive=true* parameter that is used to indicate that the configuration should be created but not enabled. Note the *return_records=true* query parameter is used to obtain the newly created key-manager keystore configuration.

```
# The API:
POST /api/security/azure-key-vaults

# The call:
curl -X POST 'https://<mgmt-ip>/api/security/azure-key-
vaults?create_inactive=true&return_records=true' -H 'accept:
application/hal+json' -d "{ \"svm\": { \"uuid\": \"4f7abf4c-9a07-11ea-
8d52-005056bbeba5\" }, \"configuration\": {\"name\": \"myConfiguration\"},
\"client_id\": \"client1\", \"tenant_id\": \"tenant1\", \"name\":
\"https:://mykeyvault.azure.vault.net/\", \"key_id\": \"https://keyvault-
test.vault.azure.net/keys/key1/a8e619fd8f234db3b0b95c59540e2a74\",
\"client_secret\" : \"myclientPwd\" }"

# The response:
{
"num_records": 1,
"records": [
  {
    "uuid": "85619643-9a06-11ea-8d52-005056bbeba5",
    "configuration": {"name": "myConfiguration"},
    "client_id": "client1",
    "tenant_id": "tenant1",
    "name": "https:://mykeyvault.azure.vault.net/",
    "key_id": "https://keyvault-
test.vault.azure.net/keys/key1/a8e619fd8f234db3b0b95c59540e2a74",
    "_links": {
      "self": {
        "href": "/api/security/azure-key-vaults/85619643-9a06-11ea-8d52-
005056bbeba5"
      }
    }
  }
]
}
```

Retrieving the AKVs configured for all clusters and SVMs

The following example shows how to retrieve all configured AKVs along with their configurations.

```
# The API:
GET /api/security/azure-key-vaults

# The call:
```

```
curl -X GET 'https://<mgmt-ip>/api/security/azure-key-vaults?fields=*'

# The response:
{
"records": [
  {
    "configuration": {
      "uuid": "024cd3cf-9a08-11ea-8d52-005056bbeba5",
      "name" : "default"
    },
    "uuid": "024cd3cf-9a08-11ea-8d52-005056bbeba5",
    "scope": "svm",
    "svm": {
      "uuid": "4f7abf4c-9a07-11ea-8d52-005056bbeba5",
      "name": "vs0"
    },
    "client_id": "client1",
    "tenant_id": "tenant1",
    "name": "https:://mykeyvault.azure.vault.net/",
    "key_id": "https://keyvault-
test.vault.azure.net/keys/key1/a8e619fd8f234db3b0b95c59540e2a74",
    "authentication_method": "client_secret",
    "enabled": true,
    "_links": {
      "self": {
        "href": "/api/security/azure-key-vaults/024cd3cf-9a08-11ea-8d52-
005056bbeba5"
      }
    }
  },
  {
    "configuration": {
      "uuid": "85619643-9a06-11ea-8d52-005056bbeba5",
      "name" : "new-config"
    },
    "uuid": "85619643-9a06-11ea-8d52-005056bbeba5",
    "scope": "cluster",
    "client_id": "client1",
    "tenant_id": "tenant1",
    "name": "https:://mykeyvault.azure.vault.net/",
    "key_id": "https://keyvault-
test.vault.azure.net/keys/key1/a8e619fd8f234db3b0b95c59540e2a74",
    "authentication_method": "certificate",
    "enabled": false,
    "_links": {
      "self": {
```

```
        "href": "/api/security/azure-key-vaults/85619643-9a06-11ea-8d52-
005056bbeba5"
      }
    }
  }
],
"num_records": 2,
"_links": {
  "self": {
    "href": "/api/security/azure-key-vaults?fields=*"
  }
}
}
```

Retrieving a specific AKV configuration

The following example retrieves a specific AKV configuration.

```
# The API:
GET /api/security/azure-key-vaults

# The call:
curl -X GET 'https://<mgmt-ip>/api/security/azure-key-vaults/85619643-
9a06-11ea-8d52-005056bbeba5?fields=*'

# The response:
{
"configuration": {
  "uuid": "85619643-9a06-11ea-8d52-005056bbeba5",
  "name" : "default"
},
"uuid": "85619643-9a06-11ea-8d52-005056bbeba5",
"scope": "cluster",
"client_id": "client1",
"tenant_id": "tenant1",
"name": "https:://mykeyvault.azure.vault.net/",
"key_id": "https://keyvault-
test.vault.azure.net/keys/key1/a8e619fd8f234db3b0b95c59540e2a74",
"authentication_method": "client_secret",
"enabled": true,
"_links": {
  "self": {
    "href": "/api/security/azure-key-vaults/85619643-9a06-11ea-8d52-
005056bbeba5"
  }
}
}
```

Retrieving the advanced properties of a specific, enabled AKV configuration

The following example retrieves the advanced properties of a specific enabled AKV configuration (inactive AKV configurations do not have these advanced properties).

```
# The API:
GET /api/security/azure-key-vaults

# The call:
curl -X GET 'https://<mgmt-ip>/api/security/azure-key-vaults/85619643-
9a06-11ea-8d52-
005056bbeba5?fields=state,azure_reachability,ekmip_reachability"'
{
```

```
"configuration": {
  "uuid": "85619643-9a06-11ea-8d52-005056bbeba5",
  "name" : "default"
},
"uuid": "85619643-9a06-11ea-8d52-005056bbeba5",
"name": "https://10.234.237.18",
"state": {
  "cluster_state": true,
  "message": "",
  "code": "0"
},
"azure_reachability": {
  "reachable": true,
  "message": "",
  "code": "0"
},
"ekmip_reachability": [
  {
    "node": {
        "uuid": "d208115f-7721-11eb-bf83-005056bb150e",
        "name": "node1",
        "_links": {
            "self": {
            "href": "/api/cluster/nodes/d208115f-7721-11eb-bf83-
005056bb150e"
            }
        }
    },
    "reachable": true,
    "message": "",
    "code": "0"
  },
  {
    "node": {
        "uuid": "e208115f-7721-11eb-bf83-005056bb150e",
        "name": "node2",
        "_links": {
            "self": {
            "href": "/api/cluster/nodes/e208115f-7721-11eb-bf83-
005056bb150e"
            }
        }
    },
    "reachable": true,
    "message": "",
    "code": "0"
```

```
  }
],
"_links": {
  "self": {
    "href": "/api/security/azure-key-vaults/85619643-9a06-11ea-8d52-
005056bbeba5"
  }
}
}
```


Updating the client secret of a specific AKV configuration

The following example updates the client secret of a specific AKV configuration.

```
# The API:
PATCH /api/security/azure-key-vaults

# The call:
curl -X PATCH 'https://<mgmt-ip>/api/security/azure-key-vaults/85619643-
9a06-11ea-8d52-005056bbeba5' -d "{ \"client_secret\": \"newSecret\" }"
```

Updating the client certificate and key of a specific AKV configuration

The following example updates the client certificate and key of a configured AKV for a specific AKV configuration.

```
# The API:
PATCH /api/security/azure-key-vaults

# The call:
curl -X PATCH 'https://<mgmt-ip>/api/security/azure-key-vaults/85619643-
9a06-11ea-8d52-005056bbeba5' -d "{ \"client_certificate\":
\"MIIQKQIBAzCCD+8GCSqGSIb3DQEHAaCCD+AEgg/cMIIP2DCCBg8GCSqGSIb3DQEHBqCCBgAw
ggX8AgEAMIIF9QYJKoZIhvcNAQcBMBwGCiqGSIb3DQEMAQYwDgQIWkY7ojViJDYCAggAgIIFyJ
PjIfmM6yTCKVw5ep2oZLwwvRca8pKhISVjw+WjWngh/f6Py/Ty0CwCjDFUZPsUUdSmk78E7SAz
0CpQyBwmUuFJQShjZjftHLKRWld3O4sJKB8DzH9Yw1C7En94cyJ1rT4WYoVFmeJcmOXx6h+NFH
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```

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```
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QU+YFhgKEYjfXN/cL70yRrJSHFgUwECHeCTQnUEU0BAgIIAA==\" }"
```

Deleting a specific AKV configuration

The following example deletes a specific, enabled AKV.

```
# The API:
DELETE /api/security/azure-key-vaults

# The call:
curl -X DELETE 'https://<mgmt-ip>/api/security/azure-key-vaults/85619643-
9a06-11ea-8d52-005056bbeba5'
```

Restoring the keys for a specific AKV configuration

The following example restores all the keys of a specific AKV configuration.

```
# The API:
POST security/azure-key-vaults/{uuid}/restore

# The call:
curl -X POST 'https://<mgmt-ip>/api/security/azure-key-vaults/85619643-
9a06-11ea-8d52-005056bbeba5/restore'

# The response:
{
"job": {
  "uuid": "6ab6946f-9a0c-11ea-8d52-005056bbeba5",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/6ab6946f-9a0c-11ea-8d52-005056bbeba5"
    }
  }
}
}
```

Rekeying the internal key for a specific AKV configuration

The following example rekeys the internal key of a specific AKV configuration.

```
# The API:
POST security/azure-key-vaults/{uuid}/rekey-internal

# The call:
curl -X POST 'https://<mgmt-ip>/api/security/azure-key-vaults/85619643-
9a06-11ea-8d52-005056bbeba5/rekey-internal'

# The response:
{
"job": {
  "uuid": "6ab6946f-9a0c-11ea-8d52-005056bbeba5",
  "_links": {
    "self": {
      "href": "/api/cluster/jobs/6ab6946f-9a0c-11ea-8d52-005056bbeba5"
    }
  }
}
}
```

Retrieve AKVs configured for all clusters and SVMs

GET `/security/azure-key-vaults`

Introduced In: 9.8

Retrieves AKVs configured for all clusters and SVMs.

Related ONTAP commands

- `security key-manager external azure show`
- `security key-manager external azure check`

Parameters

Name	Type	In	Required	Description
verify_host	boolean	query	False	Filter by verify_host • Introduced in: 9.14
enabled	boolean	query	False	Filter by enabled • Introduced in: 9.14
client_id	string	query	False	Filter by client_id
configuration.name	string	query	False	Filter by configuration.name • Introduced in: 9.14
configuration.uuid	string	query	False	Filter by configuration.uuid • Introduced in: 9.14
tenant_id	string	query	False	Filter by tenant_id
svm.uuid	string	query	False	Filter by svm.uuid
svm.name	string	query	False	Filter by svm.name

Name	Type	In	Required	Description
ekmip_reachability.message	string	query	False	Filter by ekmip_reachability.message
ekmip_reachability.node.name	string	query	False	Filter by ekmip_reachability.node.name
ekmip_reachability.node.uuid	string	query	False	Filter by ekmip_reachability.node.uuid
ekmip_reachability.reachable	boolean	query	False	Filter by ekmip_reachability.reachable
ekmip_reachability.code	string	query	False	Filter by ekmip_reachability.code
vault_host	string	query	False	Filter by vault_host • Introduced in: 9.14
state.code	string	query	False	Filter by state.code
state.available	boolean	query	False	Filter by state.available
state.message	string	query	False	Filter by state.message
uuid	string	query	False	Filter by uuid
proxy_port	integer	query	False	Filter by proxy_port
proxy_username	string	query	False	Filter by proxy_username
proxy_type	string	query	False	Filter by proxy_type
proxy_host	string	query	False	Filter by proxy_host

Name	Type	In	Required	Description
authentication_method	string	query	False	Filter by authentication_method • Introduced in: 9.10
azure_reachability.code	string	query	False	Filter by azure_reachability.code
azure_reachability.reachable	boolean	query	False	Filter by azure_reachability.reachable
azure_reachability.message	string	query	False	Filter by azure_reachability.message
scope	string	query	False	Filter by scope
verify_ip	boolean	query	False	Filter by verify_ip • Introduced in: 9.14
key_id	string	query	False	Filter by key_id
port	integer	query	False	Filter by port • Introduced in: 9.14
oauth_host	string	query	False	Filter by oauth_host • Introduced in: 9.14
name	string	query	False	Filter by name
fields	array[string]	query	False	Specify the fields to return.
max_records	integer	query	False	Limit the number of records returned.

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. • Default value: 1 • Max value: 120 • Min value: 0
return_records	boolean	query	False	The default is true for GET calls. When set to false, only the number of records is returned. • Default value: 1
order_by	array[string]	query	False	Order results by specified fields and optional [asc

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
num_records	integer	Number of records
records	array[azure_key_vault]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "authentication_method": "client_secret",
    "azure_reachability": {
      "code": "346758",
      "message": "AKV service is not reachable from all nodes -
reason."
    },
    "client_certificate":
"MIIQKQIBAzCCD+8GCSqGSIb3DQEHAaCCD+AEgg/cMIIP2DCCBg8GCSqGSIb3DQEHBqCCBg
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```

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IS+pVpfz+Mg7kl3WmkUrgF3yjTH5/C51uaSzK2KeEVoWPx/Ps2CX7ATo6AsETp8Na38dT6d
+Dm4WM4sBieKt/yOEFhiBNkgpVKAqawKRvLW3U73OIKC8VLFhhnU+ogGxcUq5mZXvMbNDIa
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```
sUYIwXgkfrEVYujCO0T5dSkk5j10X7WlDm4DHZVLJH+GtL6v9A6xFJNDQfQF0hS+wlXkTkM
q7pUiX+Qohf8QRJZEyU5VWo2CesR63j1MFpkB3xybpbjt8oI47XC20GEn3uCjwMwq/3K4ib
Hnqi16pPPRgI/u3R9TVfvOC2e0xgllrFG6cKUfogUaXoxHqP1KKjUw23bpd9L09LzSDdSHc
oDPokWzDee0ZP/Z6VH3rdjQR71kw4VBeT8nKfLP2dGBd0tpWDQhCFK7I9axxxthnv0v09x/
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oFAAQU+YFhgKEYjfXN/cL70yRrJSHFgUwECHeCTQnUEU0BAgIIAA==",
    "client_id": "aaaaaaaa-bbbb-aaaa-bbbb-aaaaaaaaaaaa",
    "client_secret": "abcdef",
    "configuration": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "default",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563434"
    },
    "ekmip_reachability": {
      "code": "346758",
      "message": "embedded KMIP server status unavailable on node.",
      "node": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "node1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "key_id":
"https://keyvault1.vault.azure.net/keys/key1/1234567890123456789012345
789012",
    "name": "https://kmip-akv-keyvault.vault.azure.net/",
    "oauth_host": "login.microsoftonline.com",
    "port": 443,
    "proxy_host": "proxy.eng.com",
    "proxy_password": "proxypassword",
    "proxy_port": 1234,
    "proxy_type": "http",
    "proxy_username": "proxyuser",
```

```
    "scope": "svm",
    "state": {
      "code": "346758",
      "message": "Top-level internal key protection key (KEK) is
unavailable on the following nodes with the associated reasons: Node:
node1. Reason: No volumes created yet for the SVM. Wrapped KEK status
will be available after creating encrypted volumes."
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "tenant_id": "zzzzzzzz-yyyy-zzzz-yyyy-zzzzzzzzzzzz",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
    "vault_host": "vault.azure.net",
    "verify_host": "",
    "verify_ip": ""
  }
}
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
next	href	
self	href	

_links

Name	Type	Description
self	href	

azure_reachability

Indicates whether or not the AKV service is reachable from all the nodes in the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
code	string	Code corresponding to the status message. Returns a 0 if AKV service is reachable from all nodes in the cluster.
message	string	Error message set when reachability is false.
reachable	boolean	Set to true when the AKV service is reachable from all nodes of the cluster.

configuration

Security keystore object reference.

Name	Type	Description
_links	_links	
name	string	Name of the configuration.

Name	Type	Description
uuid	string	Keystore UUID.

node

Name	Type	Description
_links	_links	
name	string	
uuid	string	

ekmip_reachability

Provides the connectivity status for the given SVM on the given node to all EKMIP servers configured on all nodes of the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
code	string	Code corresponding to the error message. Returns a 0 if a given SVM is able to communicate to the EKMIP servers of all of the nodes in the cluster.
message	string	Error message set when cluster-wide EKMIP server availability from the given SVM and node is false.
node	node	
reachable	boolean	Set to true if the given SVM on the given node is able to communicate to all EKMIP servers configured on all nodes in the cluster.

state

Indicates whether or not the AKV wrapped internal key is available cluster wide. This is an advanced property; there is an added computationl cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
available	boolean	Set to true when an AKV wrapped internal key is present on all nodes of the cluster.
code	string	Code corresponding to the status message. Returns a 0 if AKV wrapped key is available on all nodes in the cluster.
message	string	Error message set when top-level internal key protection key (KEK) availability on cluster is false.

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM. This field cannot be specified in a PATCH method.
uuid	string	The unique identifier of the SVM. This field cannot be specified in a PATCH method.

azure_key_vault

Name	Type	Description
_links	_links	
authentication_method	string	Authentication method for the AKV instance.

Name	Type	Description
azure_reachability	azure_reachability	Indicates whether or not the AKV service is reachable from all the nodes in the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.
client_certificate	string	PKCS12 Certificate used by the application to prove its identity to AKV.
client_id	string	Application client ID of the deployed Azure application with appropriate access to an AKV.
client_secret	string	Secret used by the application to prove its identity to AKV.
configuration	configuration	Security keystore object reference.
ekmip_reachability	array[ekmip_reachability]	
enabled	boolean	Indicates whether the configuration is enabled.
key_id	string	Key Identifier of AKV key encryption key.
name	string	Name of the deployed AKV that will be used by ONTAP for storing keys. • example: https://kmip-akv-keyvault.vault.azure.net/ • format: uri • Introduced in: 9.8 • readCreate: 1 • x-nullable: true

Name	Type	Description
oauth_host	string	Open authorization server host name.
port	integer	Authorization server and vault port number.
proxy_host	string	Proxy host.
proxy_password	string	Proxy password. Password is not audited.
proxy_port	integer	Proxy port.
proxy_type	string	Type of proxy.
proxy_username	string	Proxy username.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".
state	state	Indicates whether or not the AKV wrapped internal key is available cluster wide. This is an advanced property; there is an added computationl cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.
svm	svm	SVM, applies only to SVM-scoped objects.
tenant_id	string	Directory (tenant) ID of the deployed Azure application with appropriate access to an AKV.
uuid	string	A unique identifier for the Azure Key Vault (AKV).
vault_host	string	AKV host subdomain.

Name	Type	Description
verify_host	boolean	Verify the identity of the AKV host name.
verify_ip	boolean	Verify the identity of the AKV IP address.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Create an AKV configuration for all clusters and SVMs

POST `/security/azure-key-vaults`

Introduced In: 9.8

Configures the AKV configuration for all clusters and SVMs.

Required properties:

- `svm.uuid` or `svm.name` - Existing SVM in which to create a AKV.
- `client_id` - Application (client) ID of the deployed Azure application with appropriate access to an AKV.
- `tenant_id` - Directory (tenant) ID of the deployed Azure application with appropriate access to an AKV.
- `client_secret` or `client_certificate` - Secret or PKCS12 Certificate used by the application to prove its identity to AKV.
- `key_id`- Key Identifier of AKV encryption key.

- `name` - Name of the deployed AKV used by ONTAP for storing keys.
- `configuration.name` - The configuration name is required if and only if the `create_inactive` flag is set to true.

Optional properties:

- `port` - Authorization server and vault port number.
- `oauth_host` - Open authorization server host name.
- `vault_host` - AKV host subdomain.
- `proxy_type` - Type of proxy (http, https etc.) if proxy configuration is used.
- `proxy_host` - Proxy hostname if proxy configuration is used.
- `proxy_port` - Proxy port number if proxy configuration is used.
- `proxy_username` - Proxy username if proxy configuration is used.
- `proxy_password` - Proxy password if proxy configuration is used.

Optional parameters:

- `create_inactive` - Create an AKV configuration without enabling it. This flag is set to "false" by default.

Related ONTAP commands

- `security key-manager external azure enable`
- `security key-manager external azure create-config`
- `security key-manager external azure update-config`

Parameters

Name	Type	In	Required	Description
return_records	boolean	query	False	The default is false. If set to true, the records are returned. • Default value:

Request Body

Name	Type	Description
_links	_links	
authentication_method	string	Authentication method for the AKV instance.

Name	Type	Description
azure_reachability	azure_reachability	Indicates whether or not the AKV service is reachable from all the nodes in the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.
client_certificate	string	PKCS12 Certificate used by the application to prove its identity to AKV.
client_id	string	Application client ID of the deployed Azure application with appropriate access to an AKV.
client_secret	string	Secret used by the application to prove its identity to AKV.
configuration	configuration	Security keystore object reference.
ekmip_reachability	array[ekmip_reachability]	
enabled	boolean	Indicates whether the configuration is enabled.
key_id	string	Key Identifier of AKV key encryption key.
name	string	Name of the deployed AKV that will be used by ONTAP for storing keys. • example: https://kmip-akv-keyvault.vault.azure.net/ • format: uri • Introduced in: 9.8 • readCreate: 1 • x-nullable: true
oauth_host	string	Open authorization server host name.

Name	Type	Description
port	integer	Authorization server and vault port number.
proxy_host	string	Proxy host.
proxy_password	string	Proxy password. Password is not audited.
proxy_port	integer	Proxy port.
proxy_type	string	Type of proxy.
proxy_username	string	Proxy username.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".
state	state	Indicates whether or not the AKV wrapped internal key is available cluster wide. This is an advanced property; there is an added computationl cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.
svm	svm	SVM, applies only to SVM-scoped objects.
tenant_id	string	Directory (tenant) ID of the deployed Azure application with appropriate access to an AKV.
uuid	string	A unique identifier for the Azure Key Vault (AKV).
vault_host	string	AKV host subdomain.
verify_host	boolean	Verify the identity of the AKV host name.

Name	Type	Description
verify_ip	boolean	Verify the identity of the AKV IP address.

Example request


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication_method": "client_secret",
  "azure_reachability": {
    "code": "346758",
    "message": "AKV service is not reachable from all nodes - reason."
  },
  "client_certificate":
"MIIQKQIBAzCCD+8GCSqGSIb3DQEHAaCCD+AEgg/cMIIP2DCCBg8GCSqGSIb3DQEHBqCCBg
AwggX8AgEAMIIF9QYJKoZIhvcNAQcBMBwGCiqGSIb3DQEMAQYwDgQIWkY7ojViJDYCAggAg
IIFyJPjIfmM6yTCKVw5ep2oZLwwvRca8pKhISVjw+WjWngh/f6Py/Ty0CwCjDFUZPsUUdSm
k78E7SAz0CpQyBwmUuFJQShjZjftHLKRWld3O4sJKB8DzH9Yw1C7En94cyJ1rT4WYoVFmeJ
cmOXx6h+NFHc7njtXVsKwxc5BF88K3+3kHdV3WyVdXoeXe7yY/+EjFfjtBryp8ljuielX/N
Flh5kowhoj+yxnO0c1/0OI1iV3mTIOTXD8qrZVp9ZhAxSTRBd5uDyWMfppqxW2L+9vCUU+Z
gmRxtU3VsRLOp/T140OP7Sn1Ch2OE0bIrbYYtcpi04QcUtfEJBMlbbTbJPHDAtiO2KIQKvi
ZL4QMZgho9NNgL4MUpIbNSzDCbuIC+nNMXfgfs0nPZewY+b43H/tMmnZ8Q4kiCFwrUqbFbf
lBiPMOaJsS0eQaJhDmzM90QEgbesHWgPreAcfMUcN1+BaqHFLHUxLXDxQix6zYiCAtDX6/E
KlirRh1TFpmFX2PBd+X6uODhmwm4ub9RKj3In8t5qgtN4q/mTBXjAVDAbTIIEgobBRaXGSS
XCBc9W/jRed0DRZD9Bm8T/nV39sZNducwZa5ojYTX8fFMA0cfY6IFivXHjB00coHEEGdgCf
C0G8vACqLbb+2NuhMJPtR7Ig50iAPUMc670Z5ItOTQhyYOZ/KagOtvV8sKPCzeAkcMoHlsm
l89V79zt1fCJQTVWnaGiMj5Orcbskk6vCxhDGeU6q1kgvXJKXOYRF8/wIpv8Y7/rEpnGwE/
I0ZOXzdIDHXqA53B1zyOVem25ezWCD+kpoH89XJssYlNjIMJhjVRED61w/DbSXg2yFu/v3c
kGapVvTuyAiz5hWUNfl3ptda6GoekKnLqtL4G/RGXCnebLbXg838dlTGBznoCwGTVxXDeVY
afz8AjI10qYtTMcbN56ya9kK7IHSkrnFX24xQRQOfmD0Vob71pjdz8r1aXKvD/1X2TkYJHo
eEHq0nWpU8vwDG/xhv4YgKJGN9qsEZgiTXETUh5gak8e1tGNkP+fum+1OqlO5oS+SwNa5/e
B8eFeJl2Oi48Xi5UapaTRHPFp6kZfPXOu9cEjhILowRIi6glg7FUbmoJcu5OvDIyP9JlyQk
lw2VtgNlm1QOIvzRenXmy18XnP50NTxx2cIwby8tIcdSn2C2qhj8Gk7q8oxVZGiBgtz4Bwy
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EsO6dPLnmh4eoWidgZi9N/SoBy1aT0JpIQ6z6N5ImPfDWu9Y6TWXUg1iyOIXGsxIQVIgUNo
B5Ru/ApDxpYpFLk0fH9k9OnEWK5Im33puOQKLno1uwrOmdbG8+x1EY8wc9FvkHGH0Zh4Hyd
iCVUcYSdiGWUxVmgm4OgyiYzcpB+Ar2dzikGc4pBg8fa1a1HN5Q3TK3w4h/HeOUlmA4vWOY
uVO1H93ILGP6PWfkug+1Tam6+8yD0W5meiZ0UIZR8TF/9gDb4+4wTFnPwgfTrggEauA8tt8
uJtiyBCrYexgZTXIZGTUj/86KXQaJKCreRr/kqwJOWqkNW4CGUVzw7LiI+sArOZqUp/Tsxn
bNC73XCMNlPsnByb2zCeK13V26Crl84U9sDuqQTJRaIse01MN9AAjpa2QWEwggnBBgkqhki
G9w0BBwGgggmyBIIJrjCCCaowggmmBgsqhkiG9w0BDAoBAqCCCW4wgglqMBwGCiqGSIb3DQ
EMAQMwDgQIEjm88b1+pnkCAggABIIJSDD3P+vnllSo1mQvmYgZVfV37T3KpurJvMxQScPva
```

```
lWiF7Q1Iwasf/+N0hKKNr2j/aGZLunLkaG6mLPeBP2l2LCwnUxDu5kYffVVE90WX/bXewbY
QribwFNkNhUrSgen8BfhnRlvDrzbBLoHIvDrUFszSVBCYh31Vwgu8p9SjC8K/XlumcLdjSF
ko85XpoK23euhowjWH+X0kRoYGzorcdNE8z03BKvfR61W2XWzTSaWQ6eZHGs6Urnx5Fe/w5
0U9tMIi3BCCCqgapUHVdmHqKkmWLikX8LssUcN30JVekM2aJ9v4YO6CoegKAMVDs0tVSOv3
KbGC3GNX6lgHu4y1LOZPlPLfPXb0wDHqavlxK3zpHl8sIRzuX3HXSdEdenHYAkSV/IQZ89h
+CZUkf0nu/og8eoA8ATDA5g7fj3HXpQ6cYdrUBaHc7ruxHOiWR0GcT4XK4TTz7zZTO1wWPV
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```

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  "client_id": "aaaaaaaa-bbbb-aaaa-bbbb-aaaaaaaaaaaa",
  "client_secret": "abcdef",
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        "href": "/api/resourcelink"
      }
    },
    "name": "default",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563434"
  },
  "ekmip_reachability": {
    "code": "346758",
    "message": "embedded KMIP server status unavailable on node.",
    "node": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "node1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    }
  },
  "key_id":
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  "name": "https://kmip-akv-keyvault.vault.azure.net/",
  "oauth_host": "login.microsoftonline.com",
  "port": 443,
  "proxy_host": "proxy.eng.com",
  "proxy_password": "proxypassword",
  "proxy_port": 1234,
  "proxy_type": "http",
  "proxy_username": "proxyuser",
  "scope": "svm",
  "state": {
    "code": "346758",
    "message": "Top-level internal key protection key (KEK) is
unavailable on the following nodes with the associated reasons: Node:
node1. Reason: No volumes created yet for the SVM. Wrapped KEK status
will be available after creating encrypted volumes."
  },
  "svm": {
    "_links": {
      "self": {
```

```
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "tenant_id": "zzzzzzzz-yyyy-zzzz-yyyy-zzzzzzzzzzzz",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
    "vault_host": "vault.azure.net",
    "verify_host": "",
    "verify_ip": ""
  }
```


Response

```
Status: 201, Created
```

Name	Type	Description
_links	_links	
num_records	integer	Number of records
records	array[azure_key_vault]	

Example response


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
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      }
    },
    "authentication_method": "client_secret",
    "azure_reachability": {
      "code": "346758",
      "message": "AKV service is not reachable from all nodes -
reason."
    },
    "client_certificate":
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    "client_id": "aaaaaaaa-bbbb-aaaa-bbbb-aaaaaaaaaaaa",
    "client_secret": "abcdef",
    "configuration": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "default",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563434"
    },
    "ekmip_reachability": {
      "code": "346758",
      "message": "embedded KMIP server status unavailable on node.",
      "node": {
        "_links": {
          "self": {
            "href": "/api/resourcelink"
          }
        },
        "name": "node1",
        "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
      }
    },
    "key_id":
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789012",
    "name": "https://kmip-akv-keyvault.vault.azure.net/",
    "oauth_host": "login.microsoftonline.com",
    "port": 443,
    "proxy_host": "proxy.eng.com",
    "proxy_password": "proxypassword",
    "proxy_port": 1234,
    "proxy_type": "http",
    "proxy_username": "proxyuser",
```

```
    "scope": "svm",
    "state": {
      "code": "346758",
      "message": "Top-level internal key protection key (KEK) is
unavailable on the following nodes with the associated reasons: Node:
node1. Reason: No volumes created yet for the SVM. Wrapped KEK status
will be available after creating encrypted volumes."
    },
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "tenant_id": "zzzzzzzz-yyyy-zzzz-yyyy-zzzzzzzzzzzz",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
    "vault_host": "vault.azure.net",
    "verify_host": "",
    "verify_ip": ""
  }
}
```


Headers

Name	Description	Type
Location	Useful for tracking the resource location	string

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
3735553	Failed to create self-signed certificate.
3735664	The specified key size is not supported in FIPS mode.
3735665	The specified hash function is not supported in FIPS mode.

Error Code	Description
3735700	The specified key size is not supported.
52559972	The certificates start date is later than the current date.
65537500	A key manager has already been configured for this SVM.
65537504	Internal error. Failed to store configuration in internal database.
65537505	One or more volume encryption keys of the given SVM are stored on a key manager configured for the admin SVM.
65537506	AKV is not supported in MetroCluster configurations.
65537512	AKV cannot be configured for the given SVM as not all nodes in the cluster can enable the Azure Key Vault feature.
65537514	Failed to check if the Azure Key Vault feature is enabled.
65537518	Failed to find an interface with Cluster role.
65537523	Invalid key ID format. Example key ID format":" "https://mykeyvault.vault.azure.net/keys/key1/a8e619fd8f234db3b0b95c59540e2a74".
65537526	Failed to enable Azure Key Vault feature.
65537567	No authentication method provided.
65537573	Invalid client certificate.
65537589	The specified configuration.name already exists on the given SVM.
65537592	The configuration.name field requires an ECV of 9.14.0 or greater.
65537593	The create_inactive flag requires an effective cluster version of 9.14.0 or greater.
65537594	The configuration.name field is required when the create_inactive flag is set to true.
65537595	The configuration.name field can only be specified when the create_inactive flag is set to true.
65538902	The configuration.name is reserved for use by the system.
65538903	The configuration.name field cannot be an empty string.

Also see the table of common errors in the Response body overview section of this documentation.

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

azure_reachability

Indicates whether or not the AKV service is reachable from all the nodes in the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
code	string	Code corresponding to the status message. Returns a 0 if AKV service is reachable from all nodes in the cluster.
message	string	Error message set when reachability is false.
reachable	boolean	Set to true when the AKV service is reachable from all nodes of the cluster.

configuration

Security keystore object reference.

Name	Type	Description
_links	_links	
name	string	Name of the configuration.
uuid	string	Keystore UUID.

node

Name	Type	Description
_links	_links	

Name	Type	Description
name	string	
uuid	string	

ekmip_reachability

Provides the connectivity status for the given SVM on the given node to all EKMIP servers configured on all nodes of the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
code	string	Code corresponding to the error message. Returns a 0 if a given SVM is able to communicate to the EKMIP servers of all of the nodes in the cluster.
message	string	Error message set when cluster-wide EKMIP server availability from the given SVM and node is false.
node	node	
reachable	boolean	Set to true if the given SVM on the given node is able to communicate to all EKMIP servers configured on all nodes in the cluster.

state

Indicates whether or not the AKV wrapped internal key is available cluster wide. This is an advanced property; there is an added computationl cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
available	boolean	Set to true when an AKV wrapped internal key is present on all nodes of the cluster.
code	string	Code corresponding to the status message. Returns a 0 if AKV wrapped key is available on all nodes in the cluster.

Name	Type	Description
message	string	Error message set when top-level internal key protection key (KEK) availability on cluster is false.

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM. This field cannot be specified in a PATCH method.
uuid	string	The unique identifier of the SVM. This field cannot be specified in a PATCH method.

azure_key_vault

Name	Type	Description
_links	_links	
authentication_method	string	Authentication method for the AKV instance.
azure_reachability	azure_reachability	Indicates whether or not the AKV service is reachable from all the nodes in the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.
client_certificate	string	PKCS12 Certificate used by the application to prove its identity to AKV.
client_id	string	Application client ID of the deployed Azure application with appropriate access to an AKV.

Name	Type	Description
client_secret	string	Secret used by the application to prove its identity to AKV.
configuration	configuration	Security keystore object reference.
ekmip_reachability	array[ekmip_reachability]	
enabled	boolean	Indicates whether the configuration is enabled.
key_id	string	Key Identifier of AKV key encryption key.
name	string	Name of the deployed AKV that will be used by ONTAP for storing keys. • example: https://kmip-akv-keyvault.vault.azure.net/ • format: uri • Introduced in: 9.8 • readCreate: 1 • x-nullable: true
oauth_host	string	Open authorization server host name.
port	integer	Authorization server and vault port number.
proxy_host	string	Proxy host.
proxy_password	string	Proxy password. Password is not audited.
proxy_port	integer	Proxy port.
proxy_type	string	Type of proxy.
proxy_username	string	Proxy username.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".

Name	Type	Description
state	state	Indicates whether or not the AKV wrapped internal key is available cluster wide. This is an advanced property; there is an added computationl cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.
svm	svm	SVM, applies only to SVM-scoped objects.
tenant_id	string	Directory (tenant) ID of the deployed Azure application with appropriate access to an AKV.
uuid	string	A unique identifier for the Azure Key Vault (AKV).
vault_host	string	AKV host subdomain.
verify_host	boolean	Verify the identity of the AKV host name.
verify_ip	boolean	Verify the identity of the AKV IP address.

_links

Name	Type	Description
next	href	
self	href	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Re-key the external key in the key hierarchy

POST `/security/azure-key-vaults/{azure_key_vault.uuid}/rekey-external`

Introduced In: 9.11

Rekeys the external key in the key hierarchy for an SVM with an AKV configuration.

Required properties

- `key_id` - Key identifier of the new AKV key encryption key.

Related ONTAP commands

- `security key-manager external azure rekey-external`

Parameters

Name	Type	In	Required	Description
azure_key_vault.uuid	string	path	True	UUID of the existing AKV configuration.

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202. • Default value: 1 • Max value: 120 • Min value: 0
return_records	boolean	query	False	The default is false. If set to true, the records are returned. • Default value:

Request Body

Name	Type	Description
key_id	string	Key identifier of the AKV key encryption key.

Example request

```
{
  "key_id":
"https://keyvault1.vault.azure.net/keys/key1/1234567890123456789012345678901 2"
}
```

Response

```
Status: 202, Accepted
```

Name	Type	Description
job	job_link	

Example response

```
{
  "job": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "string"
  }
}
```

Response

```
Status: 201, Created
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
65537120	Azure Key Vault is not configured for the given SVM.
65537547	One or more volume encryption keys for encrypted volumes of this data SVM are stored in the key manager configured for the admin SVM. Use the REST API POST method to migrate this data SVM's keys from the admin SVM's key manager to this data SVM's key manager before running the rekey operation.

Also see the table of common errors in the Response body overview section of this documentation.

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

azure_key_vault_key

Name	Type	Description
key_id	string	Key identifier of the AKV key encryption key.

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

job_link

Name	Type	Description
_links	_links	
uuid	string	The UUID of the asynchronous job that is triggered by a POST, PATCH, or DELETE operation.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message

Name	Type	Description
target	string	The target parameter that caused the error.

Delete an AKV configuration

DELETE `/security/azure-key-vaults/{uuid}`

Introduced In: 9.8

Deletes an AKV configuration.

Related ONTAP commands

- `security key-manager external azure disable`

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	AKV UUID

Response

```
Status: 200, Ok
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
65536242	One or more self-encrypting drives are assigned an authentication key.
65536243	Cannot determine authentication key presence on one or more self-encrypting drives.
65536817	Internal error. Failed to determine if key manager is safe to disable.
65536827	Internal error. Failed to determine if the given SVM has any encrypted volumes.

Error Code	Description
65536834	Internal error. Failed to get existing key-server details for the given SVM.
65536867	Volume encryption keys (VEK) for one or more encrypted volumes are stored on the key manager configured for the given SVM.
65536883	Internal error. Volume encryption key is missing for a volume.
65536884	Internal error. Volume encryption key is invalid for a volume.
65536924	Cannot remove key manager that still contains one or more NSE authentication keys.
65537120	Azure Key Vault is not configured for the given SVM.
196608080	One or more nodes in the cluster have the root volume encrypted using NVE (NetApp Volume Encryption).
196608301	Internal error. Failed to get encryption type.
196608305	NAE aggregates found in the cluster.

Also see the table of common errors in the Response body overview section of this documentation.

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Retrieve the AKV configuration for an SVM specified by the UUID

GET `/security/azure-key-vaults/{uuid}`

Introduced In: 9.8

Retrieves the AKV configuration for the SVM specified by the UUID.

Related ONTAP commands

- `security key-manager external azure show`
- `security key-manager external azure check`

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	AKV UUID
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
authentication_method	string	Authentication method for the AKV instance.
azure_reachability	azure_reachability	Indicates whether or not the AKV service is reachable from all the nodes in the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.
client_certificate	string	PKCS12 Certificate used by the application to prove its identity to AKV.
client_id	string	Application client ID of the deployed Azure application with appropriate access to an AKV.
client_secret	string	Secret used by the application to prove its identity to AKV.
configuration	configuration	Security keystore object reference.
ekmip_reachability	array[ekmip_reachability]	
enabled	boolean	Indicates whether the configuration is enabled.
key_id	string	Key Identifier of AKV key encryption key.

Name	Type	Description
name	string	Name of the deployed AKV that will be used by ONTAP for storing keys. • example: https://kmip-akv-keyvault.vault.azure.net/ • format: uri • Introduced in: 9.8 • readCreate: 1 • x-nullable: true
oauth_host	string	Open authorization server host name.
port	integer	Authorization server and vault port number.
proxy_host	string	Proxy host.
proxy_password	string	Proxy password. Password is not audited.
proxy_port	integer	Proxy port.
proxy_type	string	Type of proxy.
proxy_username	string	Proxy username.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".
state	state	Indicates whether or not the AKV wrapped internal key is available cluster wide. This is an advanced property; there is an added computationl cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
svm	svm	SVM, applies only to SVM-scoped objects.
tenant_id	string	Directory (tenant) ID of the deployed Azure application with appropriate access to an AKV.
uuid	string	A unique identifier for the Azure Key Vault (AKV).
vault_host	string	AKV host subdomain.
verify_host	boolean	Verify the identity of the AKV host name.
verify_ip	boolean	Verify the identity of the AKV IP address.

Example response


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication_method": "client_secret",
  "azure_reachability": {
    "code": "346758",
    "message": "AKV service is not reachable from all nodes - reason."
  },
  "client_certificate":
"MIIQKQIBAzCCD+8GCSqGSIb3DQEHAaCCD+AEgg/cMIIP2DCCBg8GCSqGSIb3DQEHBqCCBg
AwggX8AgEAMIIF9QYJKoZIhvcNAQcBMBwGCiqGSIb3DQEMAQYwDgQIWkY7ojViJDYCAggAg
IIFyJPjIfmM6yTCKVw5ep2oZLwwvRca8pKhISVjw+WjWngh/f6Py/Ty0CwCjDFUZPsUUdSm
k78E7SAz0CpQyBwmUuFJQShjZjftHLKRWld3O4sJKB8DzH9Yw1C7En94cyJ1rT4WYoVFmeJ
cmOXx6h+NFHc7njtXVsKwxc5BF88K3+3kHdV3WyVdXoeXe7yY/+EjFfjtBryp8ljuielX/N
Flh5kowhoj+yxnO0c1/0OI1iV3mTIOTXD8qrZVp9ZhAxSTRBd5uDyWMfppqxW2L+9vCUU+Z
gmRxtU3VsRLOp/T140OP7Sn1Ch2OE0bIrbYYtcpi04QcUtfEJBMlbbTbJPHDAtiO2KIQKvi
ZL4QMZgho9NNgL4MUpIbNSzDCbuIC+nNMXfgfs0nPZewY+b43H/tMmnZ8Q4kiCFwrUqbFbf
lBiPMOaJsS0eQaJhDmzM90QEgbesHWgPreAcfMUcN1+BaqHFLHUxLXDxQix6zYiCAtDX6/E
KlirRh1TFpmFX2PBd+X6uODhmwm4ub9RKj3In8t5qgtN4q/mTBXjAVDAbTIIEgobBRaXGSS
XCBc9W/jRed0DRZD9Bm8T/nV39sZNducwZa5ojYTX8fFMA0cfY6IFivXHjB00coHEEGdgCf
C0G8vACqLbb+2NuhMJPtR7Ig50iAPUMc670Z5ItOTQhyYOZ/KagOtvV8sKPCzeAkcMoHlsm
l89V79zt1fCJQTVWnaGiMj5Orcbskk6vCxhDGeU6q1kgvXJKXOYRF8/wIpv8Y7/rEpnGwE/
I0ZOXzdIDHXqA53B1zyOVem25ezWCD+kpoH89XJssYlNjIMJhjVRED61w/DbSXg2yFu/v3c
kGapVvTuyAiz5hWUNfl3ptda6GoekKnLqtL4G/RGXCnebLbXg838dlTGBznoCwGTVxXDeVY
afz8AjI10qYtTMcbN56ya9kK7IHSkrnFX24xQRQOfmD0Vob71pjdz8r1aXKvD/1X2TkYJHo
eEHq0nWpU8vwDG/xhv4YgKJGN9qsEZgiTXETUh5gak8e1tGNkP+fum+1OqlO5oS+SwNa5/e
B8eFeJl2Oi48Xi5UapaTRHPFp6kZfPXOu9cEjhILowRIi6glg7FUbmoJcu5OvDIyP9JlyQk
lw2VtgNlm1QOIvzRenXmy18XnP50NTxx2cIwby8tIcdSn2C2qhj8Gk7q8oxVZGiBgtz4Bwy
zyKkypwm60BBRrHpAKLw6JM5RISeZnYQfIsId0tGgb61go0RJf0sFtbuvZcSvLI+2Onj8KH
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2k0GharzAJkXnNaFGnmHIIP6wJrCCSDZwDmr7GI2R5evDlRi17QUg2sulxQV0U8zezzwIUg
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EsO6dPLnmh4eoWidgZi9N/SoBy1aT0JpIQ6z6N5ImPfDWu9Y6TWXUg1iyOIXGsxIQVIgUNo
B5Ru/ApDxpYpFLk0fH9k9OnEWK5Im33puOQKLno1uwrOmdbG8+x1EY8wc9FvkHGH0Zh4Hyd
iCVUcYSdiGWUxVmgm4OgyiYzcpB+Ar2dzikGc4pBg8fa1a1HN5Q3TK3w4h/HeOUlmA4vWOY
uVO1H93ILGP6PWfkug+1Tam6+8yD0W5meiZ0UIZR8TF/9gDb4+4wTFnPwgfTrggEauA8tt8
uJtiyBCrYexgZTXIZGTUj/86KXQaJKCreRr/kqwJOWqkNW4CGUVzw7LiI+sArOZqUp/Tsxn
bNC73XCMNlPsnByb2zCeK13V26Crl84U9sDuqQTJRaIse01MN9AAjpa2QWEwggnBBgkqhki
G9w0BBwGgggmyBIIJrjCCCaowggmmBgsqhkiG9w0BDAoBAqCCCW4wgglqMBwGCiqGSIb3DQ
EMAQMwDgQIEjm88b1+pnkCAggABIIJSDD3P+vnllSo1mQvmYgZVfV37T3KpurJvMxQScPva
```

```
lWiF7Q1Iwasf/+N0hKKNr2j/aGZLunLkaG6mLPeBP2l2LCwnUxDu5kYffVVE90WX/bXewbY
QribwFNkNhUrSgen8BfhnRlvDrzbBLoHIvDrUFszSVBCYh31Vwgu8p9SjC8K/XlumcLdjSF
ko85XpoK23euhowjWH+X0kRoYGzorcdNE8z03BKvfR61W2XWzTSaWQ6eZHGs6Urnx5Fe/w5
0U9tMIi3BCCCqgapUHVdmHqKkmWLikX8LssUcN30JVekM2aJ9v4YO6CoegKAMVDs0tVSOv3
KbGC3GNX6lgHu4y1LOZPlPLfPXb0wDHqavlxK3zpHl8sIRzuX3HXSdEdenHYAkSV/IQZ89h
+CZUkf0nu/og8eoA8ATDA5g7fj3HXpQ6cYdrUBaHc7ruxHOiWR0GcT4XK4TTz7zZTO1wWPV
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2tk8bb9dy/sJmk2ojM/D/W1YtrNL4iM6azL2kVN5eiChxCof33/RuRpXfGR8YNeJTl7bq42
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xSQdD1tCtkjGRKHSTYza3zLHbBiJTIJw4z6sw9FyTTApg66UAkNtiMa1r9nqTTNaxRWEXME
QVRLzAL2F9aqjgW65xrbYXu/J9Y/SYTcYbX2SRA/JkQ+Y8F68KOoS1pvK1p5/FcEDvprTND
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s4ED092W3tR4ZXK3fnkFyrIVtRJsB3k/2smiQ6Pc1VuKHh1yTzYjXKRQcDaY3EDP9IWFtji
UfZQoZcijMWt6YXim23m2aN2Ed8qIedikR6OjFHE4Kus/2yegTszSs5CrM7NamKWzeIeNNt
h/cTcmTGDumsGNTBAsHHSq1KYpqLi4GKLHzU7WNCQRdAcIDEvMZH/CH1mZK7bzb9z038rPf
/D5WZrcK1ttd5BjTJjj7GerS0xLkvYIklAJqurjMdWYmQtT4JAHF90/zRKqFFVpSiW074bR
Q+PfaLI5C+TwoX5lYD+R91A0qyGKIkFITa8hZFY+Up+rSuREqnpAvdAVL9/gLPF6I+5+D+s
VBsGRbw2rFVRbCHdwaTQcAVPeJJy0f/+sOs/PXoejr3siORpf8iLLYOaziGYf1EtunFcCLj
8PEOznaUyouJ+lm9YKPBSLULC/sVVy6XUArYfJfq0Ag31YXpJeWPbORxVP/VCm8d/sNjWTQ
XGN/IjNZaZuliXNgq5nRkPBKwF23ZUYG4pLGpGROLup9nLSgEbpiDmN1Gq/IHSfI/8HpG/y
RAoCdqUdre3yL/f9caj8RBBHRYbbfRxtyQ9u2vsrqo1oZ7F+Mu+kjuc9BxCMvJ7JaKwvQJc
kAkzTo6t10t6MzwiqJ7Au+2oOJ2Ukb/985+TFGS219fmqWfwisOfpuvSkjRj8vIDBBm9itK
IS+pVpfz+Mg7kl3WmkUrgF3yjTH5/C51uaSzK2KeEVoWPx/Ps2CX7ATo6AsETp8Na38dT6d
+Dm4WM4sBieKt/yOEFhiBNkgpVKAqawKRvLW3U73OIKC8VLFhhnU+ogGxcUq5mZXvMbNDIa
U2LvtmtPPo/qL0bOYu76TKc1ZX0R6AXkeImQgRPsdeXPPANtW3la585oZbYxUXRfEIeKmkc
v3eSGnPCVesbxxd1SaIJe2j7H9MbHdjYkeFQuECnUhKxg63BVPl/qAEIO5+OKBzM7ctuP8a
peGW1iHAueKzJXc5IeFS/3iwkfDLRkrgzBeNIL0IINo3CoGSvn95Z8+LhNSopyqt3uB4rQk
sUYIwXgkfrEVYujCO0T5dSkk5j10X7WlDm4DHZVLJH+GtL6v9A6xFJNDQfQF0hS+wlXkTkM
q7pUiX+Qohf8QRJZEyU5VWo2CesR63j1MFpkB3xybpbjt8oI47XC20GEn3uCjwMwq/3K4ib
Hnqi16pPPRgI/u3R9TVfvOC2e0xgllrFG6cKUfogUaXoxHqP1KKjUw23bpd9L09LzSDdSHc
oDPokWzDee0ZP/Z6VH3rdjQR71kw4VBeT8nKfLP2dGBd0tpWDQhCFK7I9axxxthnv0v09x/
J7jhyoLRt5e8lMEfrqtnMWdqjFgYVEQndthZ+9/XvfNk6f5MD8fDheMuvbNThduFSZEcZCL
lW4GWKneVji4wdBrV3aCrzAzxy0H7y7nnkyCEvac503UDtr1bk1VJIVsYfYrN2S2DPbp3H2
E8r/n6jfBilwFyp3JTJvnRqQTcYHXDieW8Njq46JO6O6wsPwKQTKMfHGxxTRJdRe5yvJD54
xvFWw1YEJ/Q2c8cr1NNXEN32e5psfIJ7o48k6bsiyXnbHKSjK781Z5h8Hc3FbUF2U2p5JqL
wcD7+bknEunsbWSC37iMk7oweF3hMhKRMm9iYJ8tpxMRcWCOt7ador+Y2fYWBsu/bwXwcRI
08TElMCMGCSqGSIb3DQEJFTEWBBRymjnjEbJmrRwh4sRnwudfSQP6KDAxMCEwCQYFKw4DAh
oFAAQU+YFhgKEYjfXN/cL70yRrJSHFgUwECHeCTQnUEU0BAgIIAA==",
```

```
  "client_id": "aaaaaaaa-bbbb-aaaa-bbbb-aaaaaaaaaaaa",
  "client_secret": "abcdef",
  "configuration": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "default",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563434"
  },
  "ekmip_reachability": {
    "code": "346758",
    "message": "embedded KMIP server status unavailable on node.",
    "node": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "node1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    }
  },
  "key_id":
"https://keyvault1.vault.azure.net/keys/key1/1234567890123456789012345678901 2",
  "name": "https://kmip-akv-keyvault.vault.azure.net/",
  "oauth_host": "login.microsoftonline.com",
  "port": 443,
  "proxy_host": "proxy.eng.com",
  "proxy_password": "proxypassword",
  "proxy_port": 1234,
  "proxy_type": "http",
  "proxy_username": "proxyuser",
  "scope": "svm",
  "state": {
    "code": "346758",
    "message": "Top-level internal key protection key (KEK) is
unavailable on the following nodes with the associated reasons: Node:
node1. Reason: No volumes created yet for the SVM. Wrapped KEK status
will be available after creating encrypted volumes."
  },
  "svm": {
    "_links": {
      "self": {
```

```
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  },
  "tenant_id": "zzzzzzzz-yyyy-zzzz-yyyy-zzzzzzzzzzzz",
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
  "vault_host": "vault.azure.net",
  "verify_host": "",
  "verify_ip": ""
}
```


Error

```
Status: Default, Error
```

Name	Type	Description
error	returned_error	

Example error


```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```


Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

azure_reachability

Indicates whether or not the AKV service is reachable from all the nodes in the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
code	string	Code corresponding to the status message. Returns a 0 if AKV service is reachable from all nodes in the cluster.
message	string	Error message set when reachability is false.
reachable	boolean	Set to true when the AKV service is reachable from all nodes of the cluster.

configuration

Security keystore object reference.

Name	Type	Description
_links	_links	
name	string	Name of the configuration.
uuid	string	Keystore UUID.

node

Name	Type	Description
_links	_links	

Name	Type	Description
name	string	
uuid	string	

ekmip_reachability

Provides the connectivity status for the given SVM on the given node to all EKMIP servers configured on all nodes of the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
code	string	Code corresponding to the error message. Returns a 0 if a given SVM is able to communicate to the EKMIP servers of all of the nodes in the cluster.
message	string	Error message set when cluster-wide EKMIP server availability from the given SVM and node is false.
node	node	
reachable	boolean	Set to true if the given SVM on the given node is able to communicate to all EKMIP servers configured on all nodes in the cluster.

state

Indicates whether or not the AKV wrapped internal key is available cluster wide. This is an advanced property; there is an added computationl cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
available	boolean	Set to true when an AKV wrapped internal key is present on all nodes of the cluster.
code	string	Code corresponding to the status message. Returns a 0 if AKV wrapped key is available on all nodes in the cluster.

Name	Type	Description
message	string	Error message set when top-level internal key protection key (KEK) availability on cluster is false.

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM. This field cannot be specified in a PATCH method.
uuid	string	The unique identifier of the SVM. This field cannot be specified in a PATCH method.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Update the AKV configuration

PATCH `/security/azure-key-vaults/{uuid}`

Introduced In: 9.8

Updates the AKV configuration.

Optional properties

- `client_secret` or `client_certificate` - New secret or new PKCS12 certificate used to prove the application's identity to the AKV.
- `proxy_type` - Type of proxy (http, https etc.) if proxy configuration is used.
- `proxy_host` - Proxy hostname if proxy configuration is used.
- `proxy_port` - Proxy port number if proxy configuration is used.
- `port` - Authorization server and vault port number.
- `oauth_host` - Open authorization server host name.
- `vault_host` - AKV host subdomain.
- `verify_host` - Verify the identity of the AKV host name.
- `verfiy_ip ` - Verify the identity of the AKV IP address.
- `proxy_username` - Proxy username if proxy configuration is used.
- `proxy_password` - Proxy password if proxy configuration is used.
- `client_id` - Application (client) ID of the deployed Azure application with appropriate access to an AKV.
- `tenant_id` - Directory (tenant) ID of the deployed Azure application with appropriate access to an AKV.

Related ONTAP commands

- `security key-manager external azure update-client-secret`
- `security key-manager external azure update-credentials`
- `security key-manager external azure update-config`

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	AKV UUID

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202. • Default value: 1 • Max value: 120 • Min value: 0

Request Body

Name	Type	Description
_links	_links	
authentication_method	string	Authentication method for the AKV instance.

Name	Type	Description
azure_reachability	azure_reachability	Indicates whether or not the AKV service is reachable from all the nodes in the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.
client_certificate	string	PKCS12 Certificate used by the application to prove its identity to AKV.
client_id	string	Application client ID of the deployed Azure application with appropriate access to an AKV.
client_secret	string	Secret used by the application to prove its identity to AKV.
configuration	configuration	Security keystore object reference.
ekmip_reachability	array[ekmip_reachability]	
enabled	boolean	Indicates whether the configuration is enabled.
key_id	string	Key Identifier of AKV key encryption key.
name	string	Name of the deployed AKV that will be used by ONTAP for storing keys. • example: https://kmip-akv-keyvault.vault.azure.net/ • format: uri • Introduced in: 9.8 • readCreate: 1 • x-nullable: true
oauth_host	string	Open authorization server host name.

Name	Type	Description
port	integer	Authorization server and vault port number.
proxy_host	string	Proxy host.
proxy_password	string	Proxy password. Password is not audited.
proxy_port	integer	Proxy port.
proxy_type	string	Type of proxy.
proxy_username	string	Proxy username.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".
state	state	Indicates whether or not the AKV wrapped internal key is available cluster wide. This is an advanced property; there is an added computationl cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.
svm	svm	SVM, applies only to SVM-scoped objects.
tenant_id	string	Directory (tenant) ID of the deployed Azure application with appropriate access to an AKV.
uuid	string	A unique identifier for the Azure Key Vault (AKV).
vault_host	string	AKV host subdomain.
verify_host	boolean	Verify the identity of the AKV host name.

Name	Type	Description
verify_ip	boolean	Verify the identity of the AKV IP address.

Example request


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "authentication_method": "client_secret",
  "azure_reachability": {
    "code": "346758",
    "message": "AKV service is not reachable from all nodes - reason."
  },
  "client_certificate":
"MIIQKQIBAzCCD+8GCSqGSIb3DQEHAaCCD+AEgg/cMIIP2DCCBg8GCSqGSIb3DQEHBqCCBg
AwggX8AgEAMIIF9QYJKoZIhvcNAQcBMBwGCiqGSIb3DQEMAQYwDgQIWkY7ojViJDYCAggAg
IIFyJPjIfmM6yTCKVw5ep2oZLwwvRca8pKhISVjw+WjWngh/f6Py/Ty0CwCjDFUZPsUUdSm
k78E7SAz0CpQyBwmUuFJQShjZjftHLKRWld3O4sJKB8DzH9Yw1C7En94cyJ1rT4WYoVFmeJ
cmOXx6h+NFHc7njtXVsKwxc5BF88K3+3kHdV3WyVdXoeXe7yY/+EjFfjtBryp8ljuielX/N
Flh5kowhoj+yxnO0c1/0OI1iV3mTIOTXD8qrZVp9ZhAxSTRBd5uDyWMfppqxW2L+9vCUU+Z
gmRxtU3VsRLOp/T140OP7Sn1Ch2OE0bIrbYYtcpi04QcUtfEJBMlbbTbJPHDAtiO2KIQKvi
ZL4QMZgho9NNgL4MUpIbNSzDCbuIC+nNMXfgfs0nPZewY+b43H/tMmnZ8Q4kiCFwrUqbFbf
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B8eFeJl2Oi48Xi5UapaTRHPFp6kZfPXOu9cEjhILowRIi6glg7FUbmoJcu5OvDIyP9JlyQk
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B5Ru/ApDxpYpFLk0fH9k9OnEWK5Im33puOQKLno1uwrOmdbG8+x1EY8wc9FvkHGH0Zh4Hyd
iCVUcYSdiGWUxVmgm4OgyiYzcpB+Ar2dzikGc4pBg8fa1a1HN5Q3TK3w4h/HeOUlmA4vWOY
uVO1H93ILGP6PWfkug+1Tam6+8yD0W5meiZ0UIZR8TF/9gDb4+4wTFnPwgfTrggEauA8tt8
uJtiyBCrYexgZTXIZGTUj/86KXQaJKCreRr/kqwJOWqkNW4CGUVzw7LiI+sArOZqUp/Tsxn
bNC73XCMNlPsnByb2zCeK13V26Crl84U9sDuqQTJRaIse01MN9AAjpa2QWEwggnBBgkqhki
G9w0BBwGgggmyBIIJrjCCCaowggmmBgsqhkiG9w0BDAoBAqCCCW4wgglqMBwGCiqGSIb3DQ
EMAQMwDgQIEjm88b1+pnkCAggABIIJSDD3P+vnllSo1mQvmYgZVfV37T3KpurJvMxQScPva
```

```
lWiF7Q1Iwasf/+N0hKKNr2j/aGZLunLkaG6mLPeBP2l2LCwnUxDu5kYffVVE90WX/bXewbY
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ko85XpoK23euhowjWH+X0kRoYGzorcdNE8z03BKvfR61W2XWzTSaWQ6eZHGs6Urnx5Fe/w5
0U9tMIi3BCCCqgapUHVdmHqKkmWLikX8LssUcN30JVekM2aJ9v4YO6CoegKAMVDs0tVSOv3
KbGC3GNX6lgHu4y1LOZPlPLfPXb0wDHqavlxK3zpHl8sIRzuX3HXSdEdenHYAkSV/IQZ89h
+CZUkf0nu/og8eoA8ATDA5g7fj3HXpQ6cYdrUBaHc7ruxHOiWR0GcT4XK4TTz7zZTO1wWPV
iprUo6ayw0dYZSG22MeDA027YirM044Ifosn9CsqnNLZoOWvA2ao4ippDoBRqv5Hv6n0I3f
OAys5nPq3jJtKQ5neqUYo0MrAkoKHo0h6zn0BfvisyB88aM9N0mPD76ykbAERq7151biKbA
2tk8bb9dy/sJmk2ojM/D/W1YtrNL4iM6azL2kVN5eiChxCof33/RuRpXfGR8YNeJTl7bq42
wL70QKDBRoG1TPcLqdVqz74oshlRspfqvZsbsUatbASBt2T0YG4zfgfGh7sb2ezyougVvzd
p77wAJ6n39dc/ZLDdYDzFkQb07984y8LlhIM1AcwFcMh43gWp6A8CJ02l74ednirSqSVOPZ
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xSQdD1tCtkjGRKHSTYza3zLHbBiJTIJw4z6sw9FyTTApg66UAkNtiMa1r9nqTTNaxRWEXME
QVRLzAL2F9aqjgW65xrbYXu/J9Y/SYTcYbX2SRA/JkQ+Y8F68KOoS1pvK1p5/FcEDvprTND
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o8YvWtI4KGaDAFb6XQhR6t6mqoq/3IP6/g//PZVEnSyUVsPLDJlLF9fiOwTbMZnaiscKv8S
GEs//B9JkKrdsRrQRZcnnPjJnJLILblRVAZGuXpSKSYVPzYmOjUx3sSeLSiPoSOcqRIJ0X3
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UfZQoZcijMWt6YXim23m2aN2Ed8qIedikR6OjFHE4Kus/2yegTszSs5CrM7NamKWzeIeNNt
h/cTcmTGDumsGNTBAsHHSq1KYpqLi4GKLHzU7WNCQRdAcIDEvMZH/CH1mZK7bzb9z038rPf
/D5WZrcK1ttd5BjTJjj7GerS0xLkvYIklAJqurjMdWYmQtT4JAHF90/zRKqFFVpSiW074bR
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8PEOznaUyouJ+lm9YKPBSLULC/sVVy6XUArYfJfq0Ag31YXpJeWPbORxVP/VCm8d/sNjWTQ
XGN/IjNZaZuliXNgq5nRkPBKwF23ZUYG4pLGpGROLup9nLSgEbpiDmN1Gq/IHSfI/8HpG/y
RAoCdqUdre3yL/f9caj8RBBHRYbbfRxtyQ9u2vsrqo1oZ7F+Mu+kjuc9BxCMvJ7JaKwvQJc
kAkzTo6t10t6MzwiqJ7Au+2oOJ2Ukb/985+TFGS219fmqWfwisOfpuvSkjRj8vIDBBm9itK
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+Dm4WM4sBieKt/yOEFhiBNkgpVKAqawKRvLW3U73OIKC8VLFhhnU+ogGxcUq5mZXvMbNDIa
U2LvtmtPPo/qL0bOYu76TKc1ZX0R6AXkeImQgRPsdeXPPANtW3la585oZbYxUXRfEIeKmkc
v3eSGnPCVesbxxd1SaIJe2j7H9MbHdjYkeFQuECnUhKxg63BVPl/qAEIO5+OKBzM7ctuP8a
peGW1iHAueKzJXc5IeFS/3iwkfDLRkrgzBeNIL0IINo3CoGSvn95Z8+LhNSopyqt3uB4rQk
sUYIwXgkfrEVYujCO0T5dSkk5j10X7WlDm4DHZVLJH+GtL6v9A6xFJNDQfQF0hS+wlXkTkM
q7pUiX+Qohf8QRJZEyU5VWo2CesR63j1MFpkB3xybpbjt8oI47XC20GEn3uCjwMwq/3K4ib
Hnqi16pPPRgI/u3R9TVfvOC2e0xgllrFG6cKUfogUaXoxHqP1KKjUw23bpd9L09LzSDdSHc
oDPokWzDee0ZP/Z6VH3rdjQR71kw4VBeT8nKfLP2dGBd0tpWDQhCFK7I9axxxthnv0v09x/
J7jhyoLRt5e8lMEfrqtnMWdqjFgYVEQndthZ+9/XvfNk6f5MD8fDheMuvbNThduFSZEcZCL
lW4GWKneVji4wdBrV3aCrzAzxy0H7y7nnkyCEvac503UDtr1bk1VJIVsYfYrN2S2DPbp3H2
E8r/n6jfBilwFyp3JTJvnRqQTcYHXDieW8Njq46JO6O6wsPwKQTKMfHGxxTRJdRe5yvJD54
xvFWw1YEJ/Q2c8cr1NNXEN32e5psfIJ7o48k6bsiyXnbHKSjK781Z5h8Hc3FbUF2U2p5JqL
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08TElMCMGCSqGSIb3DQEJFTEWBBRymjnjEbJmrRwh4sRnwudfSQP6KDAxMCEwCQYFKw4DAh
oFAAQU+YFhgKEYjfXN/cL70yRrJSHFgUwECHeCTQnUEU0BAgIIAA==",
```

```
  "client_id": "aaaaaaaa-bbbb-aaaa-bbbb-aaaaaaaaaaaa",
  "client_secret": "abcdef",
  "configuration": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "default",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563434"
  },
  "ekmip_reachability": {
    "code": "346758",
    "message": "embedded KMIP server status unavailable on node.",
    "node": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "node1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    }
  },
  "key_id":
"https://keyvault1.vault.azure.net/keys/key1/1234567890123456789012345
6789012",
  "name": "https://kmip-akv-keyvault.vault.azure.net/",
  "oauth_host": "login.microsoftonline.com",
  "port": 443,
  "proxy_host": "proxy.eng.com",
  "proxy_password": "proxypassword",
  "proxy_port": 1234,
  "proxy_type": "http",
  "proxy_username": "proxyuser",
  "scope": "svm",
  "state": {
    "code": "346758",
    "message": "Top-level internal key protection key (KEK) is
unavailable on the following nodes with the associated reasons: Node:
node1. Reason: No volumes created yet for the SVM. Wrapped KEK status
will be available after creating encrypted volumes."
  },
  "svm": {
    "_links": {
      "self": {
```

```
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    },
    "tenant_id": "zzzzzzzz-yyyy-zzzz-yyyy-zzzzzzzzzzzz",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412",
    "vault_host": "vault.azure.net",
    "verify_host": "",
    "verify_ip": ""
  }
```


Response

```
Status: 200, Ok
```

Response

```
Status: 202, Accepted
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
65537120	Azure Key Vault is not configured for the given SVM.
65537504	Internal error. Failed to store configuration in internal database.
65537517	The field "client_secret" must be specified.
65537540	Invalid client secret.
65537541	No inputs were provided for the patch request.

Error Code	Description
65537547	One or more volume encryption keys for encrypted volumes of this data SVM are stored in the key manager configured for the admin SVM. Use the REST API POST method to migrate this data SVM's keys from the admin SVM's key manager to this data SVM's key manager before running the rekey operation.
65537573	Invalid client certificate.
65537577	The AKV certificate authentication method cannot be configured for the given SVM as not all nodes in the cluster support the AKV certificate authentication.

Also see the table of common errors in the Response body overview section of this documentation.

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

azure_reachability

Indicates whether or not the AKV service is reachable from all the nodes in the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
code	string	Code corresponding to the status message. Returns a 0 if AKV service is reachable from all nodes in the cluster.
message	string	Error message set when reachability is false.
reachable	boolean	Set to true when the AKV service is reachable from all nodes of the cluster.

configuration

Security keystore object reference.

Name	Type	Description
_links	_links	
name	string	Name of the configuration.
uuid	string	Keystore UUID.

node

Name	Type	Description
_links	_links	

Name	Type	Description
name	string	
uuid	string	

ekmip_reachability

Provides the connectivity status for the given SVM on the given node to all EKMIP servers configured on all nodes of the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
code	string	Code corresponding to the error message. Returns a 0 if a given SVM is able to communicate to the EKMIP servers of all of the nodes in the cluster.
message	string	Error message set when cluster-wide EKMIP server availability from the given SVM and node is false.
node	node	
reachable	boolean	Set to true if the given SVM on the given node is able to communicate to all EKMIP servers configured on all nodes in the cluster.

state

Indicates whether or not the AKV wrapped internal key is available cluster wide. This is an advanced property; there is an added computationl cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.

Name	Type	Description
available	boolean	Set to true when an AKV wrapped internal key is present on all nodes of the cluster.
code	string	Code corresponding to the status message. Returns a 0 if AKV wrapped key is available on all nodes in the cluster.

Name	Type	Description
message	string	Error message set when top-level internal key protection key (KEK) availability on cluster is false.

svm

SVM, applies only to SVM-scoped objects.

Name	Type	Description
_links	_links	
name	string	The name of the SVM. This field cannot be specified in a PATCH method.
uuid	string	The unique identifier of the SVM. This field cannot be specified in a PATCH method.

azure_key_vault

Name	Type	Description
_links	_links	
authentication_method	string	Authentication method for the AKV instance.
azure_reachability	azure_reachability	Indicates whether or not the AKV service is reachable from all the nodes in the cluster. This is an advanced property; there is an added computational cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.
client_certificate	string	PKCS12 Certificate used by the application to prove its identity to AKV.
client_id	string	Application client ID of the deployed Azure application with appropriate access to an AKV.

Name	Type	Description
client_secret	string	Secret used by the application to prove its identity to AKV.
configuration	configuration	Security keystore object reference.
ekmip_reachability	array[ekmip_reachability]	
enabled	boolean	Indicates whether the configuration is enabled.
key_id	string	Key Identifier of AKV key encryption key.
name	string	Name of the deployed AKV that will be used by ONTAP for storing keys. • example: https://kmip-akv-keyvault.vault.azure.net/ • format: uri • Introduced in: 9.8 • readCreate: 1 • x-nullable: true
oauth_host	string	Open authorization server host name.
port	integer	Authorization server and vault port number.
proxy_host	string	Proxy host.
proxy_password	string	Proxy password. Password is not audited.
proxy_port	integer	Proxy port.
proxy_type	string	Type of proxy.
proxy_username	string	Proxy username.
scope	string	Set to "svm" for interfaces owned by an SVM. Otherwise, set to "cluster".

Name	Type	Description
state	state	Indicates whether or not the AKV wrapped internal key is available cluster wide. This is an advanced property; there is an added computationl cost to retrieving its value. The property is not populated for either a collection GET or an instance GET unless it is explicitly requested using the `fields` query parameter or GET for all advanced properties is enabled.
svm	svm	SVM, applies only to SVM-scoped objects.
tenant_id	string	Directory (tenant) ID of the deployed Azure application with appropriate access to an AKV.
uuid	string	A unique identifier for the Azure Key Vault (AKV).
vault_host	string	AKV host subdomain.
verify_host	boolean	Verify the identity of the AKV host name.
verify_ip	boolean	Verify the identity of the AKV IP address.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code

Name	Type	Description
message	string	Error message
target	string	The target parameter that caused the error.

Re-key the internal key in the key hierarchy for an SVM

POST `/security/azure-key-vaults/{uuid}/rekey-internal`

Introduced In: 9.10

Rekeys the internal key in the key hierarchy for an SVM with an AKV configuration.

Related ONTAP commands

- `security key-manager external azure rekey-internal`

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	UUID of the existing AKV configuration.

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202. • Default value: 1 • Max value: 120 • Min value: 0
return_records	boolean	query	False	The default is false. If set to true, the records are returned. • Default value:

Response

```
Status: 202, Accepted
```

Name	Type	Description
job	job_link	

Example response

```
{
  "job": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "string"
  }
}
```

Response

```
Status: 201, Created
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
65537120	Azure Key Vault is not configured for the given SVM.
65537547	One or more volume encryption keys for encrypted volumes of this data SVM are stored in the key manager configured for the admin SVM. Use the REST API POST method to migrate this data SVM's keys from the admin SVM's key manager to this data SVM's key manager before running the rekey operation.
65537559	There are no existing internal keys for the SVM. A rekey operation is allowed for an SVM with one or more encryption keys.

Also see the table of common errors in the Response body overview section of this documentation.

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

job_link

Name	Type	Description
_links	_links	
uuid	string	The UUID of the asynchronous job that is triggered by a POST, PATCH, or DELETE operation.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Restore keys for an SVM from a configured AKV

POST `/security/azure-key-vaults/{uuid}/restore`

Introduced In: 9.10

Restore the keys for an SVM from a configured AKV.

Related ONTAP commands

- `security key-manager external azure restore`

Parameters

Name	Type	In	Required	Description
uuid	string	path	True	UUID of the existing AKV configuration.
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202. • Default value: 1 • Max value: 120 • Min value: 0

Name	Type	In	Required	Description
return_records	boolean	query	False	The default is false. If set to true, the records are returned. • Default value:

Response

```
Status: 202, Accepted
```

Name	Type	Description
job	job_link	

Example response

```
{
  "job": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "uuid": "string"
  }
}
```

Response

```
Status: 201, Created
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
65537120	Azure Key Vault is not configured for the given SVM.
65537515	Failed to restore keys on some nodes in the cluster.
65537544	Missing wrapped top-level internal key protection key (KEK) from internal database.

Also see the table of common errors in the Response body overview section of this documentation.

Name	Type	Description
error	returned_error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

job_link

Name	Type	Description
_links	_links	
uuid	string	The UUID of the asynchronous job that is triggered by a POST, PATCH, or DELETE operation.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

returned_error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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